

Nguyen Huu Thang

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CAREER OBJECTIVE

Research-oriented Software Engineering student with recognized achievements in AI and Natural Language Processing. Strong foundation in Machine Learning and Deep Learning, with hands-on experience designing intelligent systems built on advanced RAG paradigms such as GraphRAG, CAG, and knowledge graph-integrated architectures. Seeking an AI or Software Engineering Internship to contribute to and learn from engineering teams.

EDUCATION

Ho Chi Minh City University of Industry and Trade – HUIT
Bachelor of Information Technology – Software Engineering | GPA: 3.0/4.0

Ho Chi Minh City, VN
Sep 2023 – Present

TECHNICAL STACKS

Programming: Python, C# (ASP.NET Core), SQL, FastAPI
AI/Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Transformers
NLP & LLM Ecosystem: BERT, RAG Pipelines, Knowledge Graph, LangChain, LLM Fine-tuning
Computer Vision: CNN, Vision Transformers, Image Classification
Databases & MLOps: PostgreSQL, MongoDB, Redis, Docker, Git

EXPERIENCE

Sigma Group
Web Collaborator

Ho Chi Minh City, VN
Nov 2025 – Jan 2026

- **Insurance Web Interface:** Developed and customized the UI for an insurance sales platform using Liquid.
- **Visual Structuring:** Translated design mockups into responsive, user-friendly web pages.

PROJECTS

ViRes-HotelABSA – Vietnamese Aspect-Based Sentiment Analysis
Research Project | PhoBERT, Knowledge Graph, CRF, Python

Paper: Link
NLP / AI Research

- **Dataset Construction:** Built a Vietnamese dataset for ABSA corpus of 10,406 hotel reviews annotated with 47,163 entity–aspect–sentiment triples using an LLM-assisted pipeline with human review achieving Cohen’s $\kappa = 0.88$ for aspect boundary identification and $\kappa = 0.97$ for sentiment polarity classification.
- **Model Architecture:** Proposed PhoBERT-KGAN-CRF combining PhoBERT with a domain KG 143k nodes, 6.38M edges via Attention-based Graph Pruning, decoded by a BIOES in the CRF.
- **Experimental Results:** Achieved aspect extraction F1 77.30% and sentiment F1 81.28%, outperforming PhoBERT, PhoBERT-KG, and PhoBERT-CRF baselines.

LexiLingo – AI Language Learning Platform

Personal Project | FastAPI, Flutter, PostgreSQL, Redis, LoRA/PEFT

GitHub: Link | Live: Link
AI / Full-Stack Development

- **System Architecture:** Built a 4-service English learning platform with FastAPI, Flutter, PostgreSQL, and Redis for lessons, vocabulary tracking, and spaced-repetition review.
- **GraphCAG Tutoring Pipeline:** Designed a 3-layer GraphCAG pipeline L0/L1/L2 that routes learner input through knowledge-graph expansion, retrieval, and response generation, improving consistency for multi-turn tutoring and grammar/vocabulary feedback.
- **Voice Practice & Model Adaptation:** Integrated Whisper STT and Piper TTS for speaking practice, and fine-tuned a 1.7B Qwen model with LoRA/PEFT for grammar correction and vocabulary feedback.

HONORS & ACTIVITIES

DASGRI 2026 Author: Presented KG-enhanced PhoBERT research for Vietnamese ABSA.

Certificate

1st Prize – Scientific Research: Built ViRes-HotelABSA dataset for sentiment analysis at HUIT.

Certificate

Data Science Finalist: Reached final round; optimized feature engineering and model evaluation.

Certificate

LANGUAGES

Vietnamese: Native

English: Technical Reading – Advanced, Spoken Communication – Intermediate